

Highly Efficient Broadband Thin-Film Lithium Niobate MZM for 1 μ m Optical Communications

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Abstract We demonstrate the first compact 1 μ m TFLN modulator with $V_{\pi} = 4.2$ V ($V_{\pi L} = 1.1$ V $\cdot$ cm) and electro-optical bandwidth > 110 GHz. We successfully demonstrate data modulation up to 145 GHz electro-optic bandwidth by upconverting a 20 GBd signal combined with single-ended heterodyne detection.

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Introduction

The most effective technique to improve efficiency, increase the modulation bandwidth and reduce the size of a modulator is to shrink the operating wavelength. For TFLN specifically, the Pockels EO effect scales inversely with wavelength, and the optical mode shrinks enabling electrodes to get closer together. Unfortunately, telecom has been centered at 1550 nm due to the combination of the Erbium doped fiber amplifier coinciding with the low loss fiber transmission window in SMF, and datacom is stuck in the O-band (1300 nm) where fiber dispersion is near zero. In SMF based communication systems, operating at 1 μ m would be problematic due to large fiber propagation losses above 1 dB/km which would limit the use case to datacom and short reach.

Operating at 1 μ m could achieve higher bandwidth and lower power communication systems for both terrestrial systems, satellite and free-space systems when combined with broadband (>12 THz) and energy efficient Ytterbium doped fiber amplifiers (YDFA) [1]. Optical satellite communication systems which have size weight and power (SWaP) constraints would benefit from both a much lower free space beam divergence at 1 μ m and higher achievable launch powers achievable using energy efficient YDFA. In fiber systems, hollow-core fibers have shown losses of 0.16 dB/km in the 1 μ m band [2] while maintaining low dispersion which would enable use of the YDFA in long haul and broadband telecom systems.

Despite its theoretical advantage, the MZMs operating at this wavelength reported as of today fail to surpass or even match the performance of their 1550 nm counterpart [3,4]. Recent work has reported a TFLN MZM at 1064 nm exhibiting good efficiency and EO bandwidth approaching 100 GHz, underscoring the potential of this wavelength range [5]. In this work, we further advance

the state of the art by realizing a compact high-bandwidth TFLN traveling-wave MZ modulator at 1064 nm that features the lowest reported $V_{\pi L}$ together with a measured EO bandwidth well beyond 110 GHz. Compared to a similar optimized design for 1550 nm in the same platform, this modulator achieves significantly higher BW for the same V_{π} and is 2x shorter.

1 μ m MZ Modulator

The 1 μ m I/Q modulators were fabricated on a TFLN commercial foundry multi-project wafer run with 350 nm of Lithium Niobate on a Silicon substrate. The design of the 1 μ m modulator differs from the one operating in C-band in its MMI, optical edge couplers, and phase shifters. The optical routing requires narrower waveguide width (< 600 nm) to ensure single mode operation. Additionally, care must be taken at these short wavelengths to avoid polarization mixing effects in bends due to the strong birefringence of TFLN [6]. To this goal, working with thin LN slabs is beneficial.

The TWEs are 2.9 mm long (Fig. 1), are loaded with periodic capacitive electrodes, and are terminated on chip. Thanks to the higher confinement of the optical mode at shorter wavelengths, it is possible to reduce the electrodes gap of the capacitive load, effectively increasing the modulation efficiency without incurring into high optical losses. The insertion loss of the I/Q device at 1050 nm is 4.5 dB, including coupling to lensed fibers and on-chip losses. The measured V_{π} is 4.2 V (Fig. 2a) which translates to a record low $V_{\pi L} = 1.1$ V $\cdot$ cm (accounting for the fill factor), about a factor 2 better than what typically reported in TFLN-MZMs operating at 1550 nm [7,8], and about 40% lower than what recently reported for fast 1 μ m MZMs [5]. While the Pockels effect's greater efficiency is the largest contributor to this reduction, the shorter wavelength's TWE optimization also significantly impacts it.

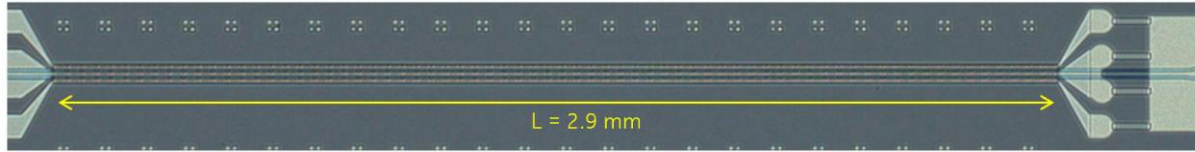


Fig. 1: microscope picture of a single arm of the compact 1 μ m TFLN I/Q-MZM

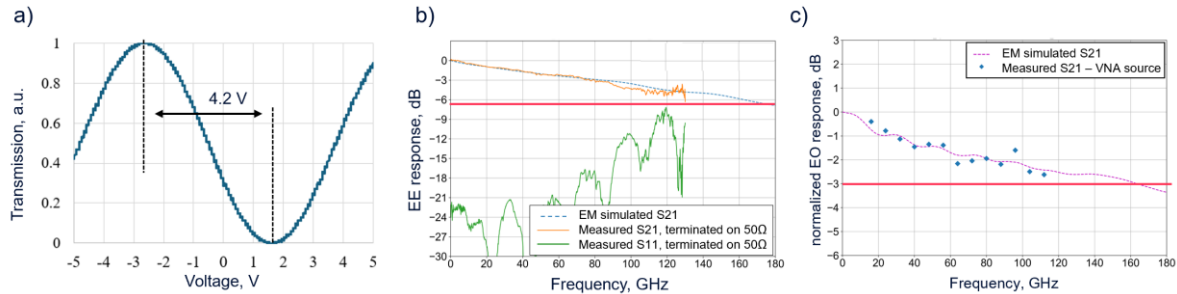


Fig. 2: a) V_{pi} measurement of the 1 μ m TFLN MZM at 1050nm, b) EE and c) EO frequency response

The high frequency response is first investigated in the electrical domain with VNA measurements of a modulator without on-chip terminations up to 125 GHz (Fig. 2b). The measured curve matches well the EM simulations and exhibits a 6.4 dB bandwidth exceeding the measurement limit of the setup. From the electrical S-parameters, the impedance of the TWE is extracted to be 50 Ω and the group delay around 2.27, which closely matches the one of the optical wave group delay at 1064 nm. Due to the lack of availability of 1 μ m photodetectors of sufficient bandwidth, the electro-optic response of the MZM is measured with an OSA by measuring the power in the modulated sidebands. For this measurement, the VNA is used as source up to 110 GHz and the power is carefully calibrated down to the probe's tip. The measured values are reported in Fig. 2c and show a 3dB-EO bandwidth well exceeding 110 GHz. From the EE measurements and the well-matching EM simulations, which include input taper, TWE, and on-chip terminations, the component's actual EO BW is expected to be close to 160 GHz. The measured response drop below 20 GHz and 20 GHz ripples are due to the on-chip terminations which were measured to be slightly higher than expected (58 Ω instead of 50 Ω).

With respect to a traditional 1550 nm modulator fabricated on the same platform (TFLN on Silicon substrate) and comparable V_{π} , our component is double as efficient and therefore shorter and able to operate at higher speed [9]. With correct terminations and shorter electrical input lines, a factor of 2 in speed is to be achieved. When coupled with recently published techniques to extend the EO bandwidth (e.g. Si substrate removal [10]), the 1 μ m MZM is believed to enable EO BWs approaching 300 GHz while keeping the V_{pi} at a reasonable low value.

Data Modulation and Detection

The experimental setup is shown in Fig. 3. The optical transmitter consists of a laser, a single arm of the I/Q MZM (due to only a single probe availability), and a high-speed RF source. The transmitter laser has a fixed wavelength of $\sim$ 1064.99 nm, an output power of 10.5 dBm and a linewidth of $\sim$ 0.5 kHz. The modulator has an insertion loss of 14.4 dB after fiber attachment. Due to the fact that our 1 μ m photodiode (PD) is a single-ended PD and has only $\sim$ 20 GHz bandwidth, we created a special banded RF signal to show the speed of the modulator while keeping the detection bandwidth low. In the RF domain, we follow the principle of digital band interleaving (DBI) method [11], upconverting a 20 GBaud signal to an intermediate frequency of 135 GHz, as shown in Fig. 3(i). In other words, we generate an RF signal that covers the baseband frequencies from 125 GHz to 145 GHz. In this way, we can use the high frequency part of the modulator to demonstrate its bandwidth, meanwhile enabling heterodyne detection with a low-bandwidth PD. Note that since we modulate with the real part of a complex valued signal a single modulator arm biased at the null point, we obtain two images, both replicas of the complex data. By placing the LO next to one of them and detecting with a narrow-band filter, only one of the two images is selected, therefore giving rise to QAM constellation diagram in the complex plane, instead of PAM or ASK. The 135-GHz-centered 20 GBaud signal has a peak-to-peak voltage of $\sim$ 1.9 V. The resulting modulated light is then amplified by an YDFA with gain of 26 dB and NF of 7.7 dB. An optical spectrum after the amplification is shown in Fig. 3(ii). The signal (the longer wavelength side band) is then filtered by an ASE filter and combined with a local oscillator (LO) via an optical coupler. The LO is a free running laser spaced $\sim$ 150 GHz away from the transmitter laser so that

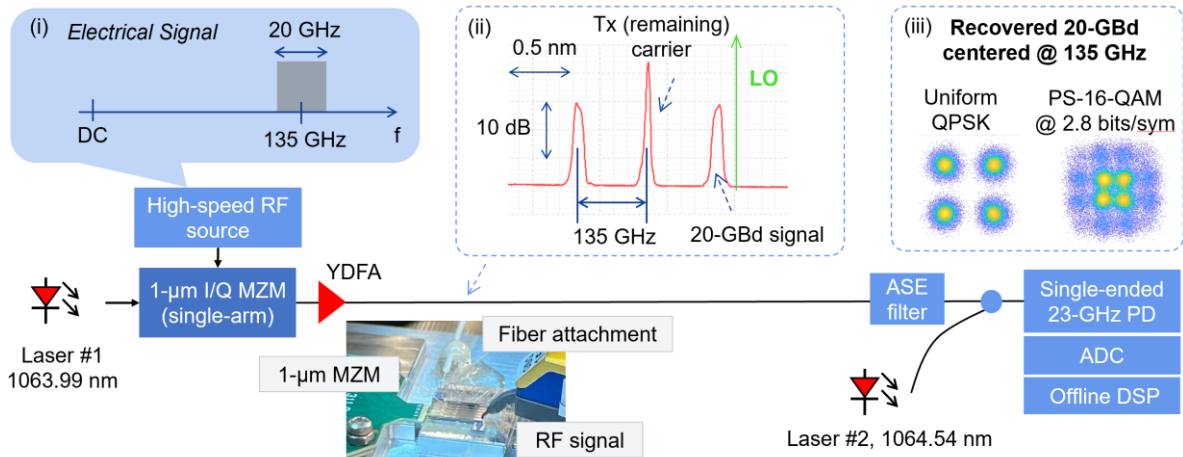


Fig. 3: Data modulation measurement setup of the 1 μ m and insets of i) RF source spectrum, ii) modulated optical spectrum after amplification, and iii) recovered constellations

we can detect one of the 135 GHz centered side bands to a < 25 GHz baseband signal. The LO has a power of 15 dBm and a linewidth of a few kHz. The down-converted signal is then sampled by a real-time scope followed by offline digital signal processing (DSP). Regarding the modulation format, we use QPSK and PS-16-QAM with root-raised-cosine (RRC) filters with a 1% spectral roll-off. We assume a rate-0.7932 (26.07% overhead) concatenated forward error correction (FEC) coding [12]. The required normalized generalized mutual information (NGMI) for the FEC is $NGMI^* = 0.845$. At the receiver, the DSP steps are i) frequency offset compensation; ii) frame synchronization; iii) equalization by a 1x1 81-tap least-mean square error (LMS) equalizer operating at 2x oversampling; iv) symbol decision. We choose to modulate a uniform QPSK and a probabilistically shaped 16-QAM at an entropy of 2.8bits/symbol. The recovered QPSK signals have a symbol signal-to-noise ratio (SNR) of 11.0 dB, with a BER of 1.4×10^{-4} and an NGMI of 0.9994. The recovered PS-16-QAM signals have a symbol SNR of 10.3 dB, with a BER of 3.7×10^{-3} and an NGMI of 0.8564. Both are above the FEC limit and can be successfully recovered error free. The recovered constellations are shown in Fig. 3(iii). The relatively low end-to-end SNR is a result of a combination of limited input optical power for the modulator and the limited common-mode rejection due to the single-ended PD coherent detection. We expect the performance to be drastically improved with proper 1 μ m optical amplifications due to increased OSNR.

Conclusion

In conclusion, we demonstrate a compact Mach-Zehnder modulator operating at 1 μ m with EO bandwidth well exceeding 110 GHz together with record high efficiency and $V_{\pi} = 4.2$ V. With all possible design optimizations (i.e., optimized terminations, substrate removal, lower loss metals)

designs could achieve 300 GHz BW at 1 μ m. We expect application of this modulator in future HCF based optical transmission systems, satellite comms, and energy efficient datacom.

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